CO4 ASSESSMENT TOOL-2(Industry Problem Solving)

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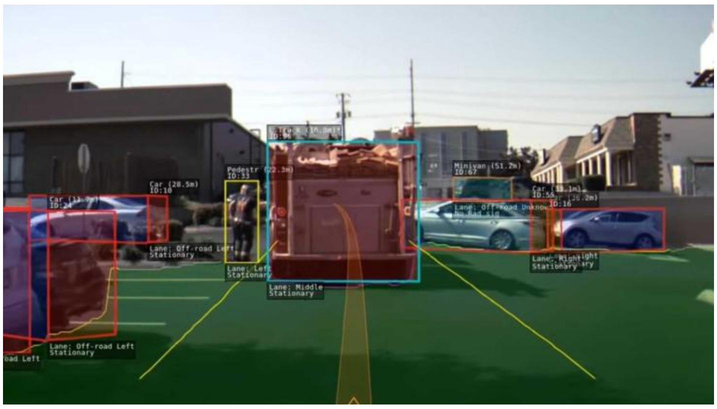
1. Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system. The system must be highly accurate and fast because incorrect or delayed object detection can create safety risks.

The given constraints are:

- Accuracy must be at least 90%
- Latency must not exceed 70 ms
- Computational requirement should be moderate, since the vehicle has moderate computational resources.



Comparison of the Three Methods

Method	Accuracy	Latency	Computation	Evaluation
Viola-Jones	82%	30 ms	Low	Fails accuracy requirement
YOLO	93%	60 ms	Moderate	Satisfies all requirements
Deep Learning	97%	90 ms	High	Fails latency and computation

1. Viola-Jones

Viola-Jones has an accuracy of 82%, which is below the required minimum of 90%. Although its latency is only 30 ms, making it very fast, and its computational requirement is low, the accuracy is not sufficient for an autonomous vehicle. A vehicle needs reliable detection of objects such as pedestrians, vehicles, and obstacles. Therefore, Viola-Jones fails the accuracy constraint and cannot be selected.

2. YOLO

YOLO provides an accuracy of 93%, which is higher than the required 90%. Its latency is 60 ms, which is also below the maximum acceptable latency of 70 ms. In addition, YOLO requires moderate computational resources, which matches the resources available in the vehicle.

Therefore:

- Accuracy: 93% ≥ 90%
- Latency: 60 ms ≤ 70 ms
- Computation: Moderate

YOLO satisfies all the given requirements. It provides a good balance between detection accuracy, processing speed, and computational cost.

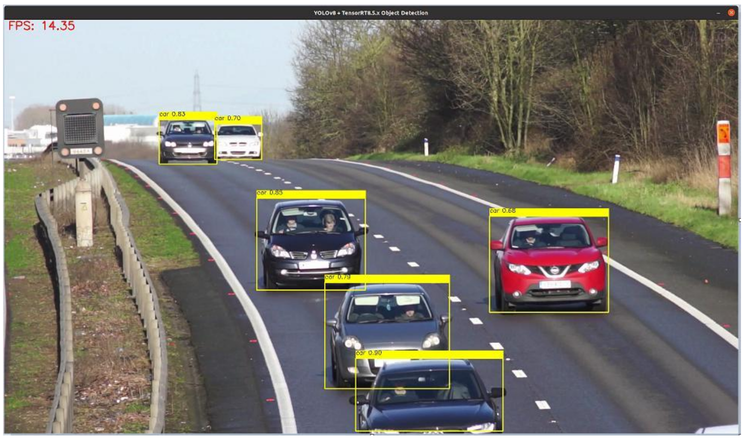
3. Deep Learning

The Deep Learning method achieves the highest accuracy of 97%, which is excellent for object detection. However, its latency is 90 ms, which is greater than the allowed maximum of 70 ms. This delay can negatively affect real-time vehicle operation because the system may take too long to detect obstacles.

It also requires high computational resources, while the vehicle only supports moderate computational resources. Hence, even though its accuracy is excellent, it fails the latency and computational requirements.

Final Decision

YOLO is the most suitable method for the autonomous vehicle vision system. It is the only method that satisfies all three constraints simultaneously. Viola-Jones is fast and computationally efficient but does not provide sufficient accuracy. Deep Learning provides the highest accuracy but has excessive latency and computational requirements.



Thus, **YOLO** should be selected because its 93% accuracy, 60 ms latency, and moderate computational requirement provide the best overall balance for real-time autonomous vehicle detection.

2. Smart Factory Product Inspection Selection

A manufacturing company needs a real-time computer vision system to identify defective products on a production line. Since the inspection happens continuously, the system must have high accuracy, low processing time, and reasonable computational requirements.

Given Constraints

- Accuracy ≥ 95% — otherwise, defective products may be missed.
- Processing time ≤ 50 ms — otherwise, production speed may be affected.
- Moderate computation — because the available hardware supports moderate computational resources.

Comparison of Methods

Method	Accuracy	Processing Time	Computation	Result
Traditional Image Analysis	91%	25 ms	Low	Fails accuracy
YOLO	96%	45 ms	Moderate	Satisfies all
Deep Learning	98%	75 ms	High	Fails time & computation

1. Traditional Image Analysis

Traditional Image Analysis achieves 91% accuracy, which is below the required 95%. Although its processing time is only 25 ms and it requires low computational resources, its lower accuracy can result in defective products being incorrectly classified as good.

Therefore:

- Accuracy: 91% < 95%
- Processing time: 25 ms ≤ 50 ms
- Computation: Low

Since accuracy is a critical requirement, this method cannot be selected.

2. YOLO

YOLO achieves 96% accuracy, which is above the required 95%. Its processing time is 45 ms, which is within the maximum limit of 50 ms. It also requires moderate computation, matching the available hardware.

Therefore:

- Accuracy: 96% ≥ 95%
- Processing time: 45 ms ≤ 50 ms
- Computation: Moderate

YOLO satisfies all three constraints. It can detect defective products accurately while processing images quickly enough to support real-time production.

3. Deep Learning

The Deep Learning method provides the highest accuracy of 98%, making it very effective at detecting defects. However, its processing time is 75 ms, which exceeds the maximum allowed 50 ms. It also requires high computational resources, while the factory hardware only supports moderate computation.

Therefore:

- Accuracy: 98% $\geq$ 95%
- Processing time: 75 ms > 50 ms
- Computation: High

Thus, despite its excellent accuracy, Deep Learning is not suitable for this real-time production environment.

Final Decision

YOLO is the best choice for the smart factory product inspection system. It is the only method that satisfies all the specified constraints. Traditional Image Analysis is fast and computationally efficient but has insufficient accuracy. Deep Learning has the highest accuracy but exceeds the processing-time limit and requires excessive computational resources.

Therefore, YOLO should be selected because its 96% accuracy, 45 ms processing time, and moderate computational requirement provide the best balance between defect-detection accuracy, real-time performance, and hardware limitations.